EPIDEMIOLOGICAL MACHINE LEARNING • POPULATION SCREENING

When Training Labels Change Who Gets Screened

Temporal evaluation of diagnosis-based versus biomarker-augmented labels in CDC NHANES under finite screening capacity

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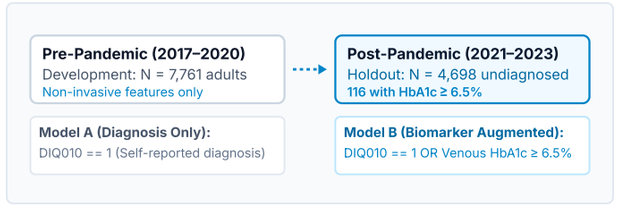
01 THE CLINICAL PROBLEM

Population screening triage algorithms frequently train on **administrative proxy labels** (e.g. billing codes, prior clinical diagnoses) because objective laboratory tests are resource-constrained.

However, clinical diagnoses require clinical encounters. When screening algorithms learn from diagnosis history, individuals facing healthcare access barriers are misclassified as healthy baselines, distorting screening queues.

02 STUDY DESIGN & TEMPORAL PROTOCOL

We evaluated whether replacing administrative labels with **biomarker-augmented labels** changes triage selection, holding non-invasive features and screening capacity constant.



Feature Space Constraint: Both models use identical non-invasive features (Age, Sex, BMI, Waist circumference, Poverty Income Ratio). Lab tests and healthcare encounters were strictly barred from model inputs.

03 DATA ENGINEERING & QUALITY NOTE

Embedded Lakehouse: Ingested 10 SAS transport (.XPT) cycles into an in-process DuckDB analytical lakehouse, executing SQL-driven cohort exclusions and exporting zero-copy Parquet files.

Data Contracts: Pandera schemas verified biological bounds (HbA1c: 2.0% to 25.0%), confirmed 0 pregnant participants, and logged stable covariate distributions across survey cycles (PSI < 0.04 for age, BMI, waist).

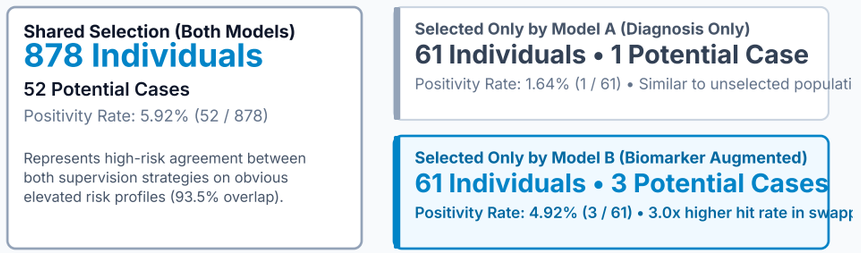
KEY FINDING

Changing training labels altered screening priorities and identified two additional potential cases at the same screening capacity.

04 PAIRED-SELECTION QUEUE ANALYSIS AT 20% CLINIC CAPACITY

Capacity K = 939 slots / 4,698 adults

At 20% fixed screening capacity, both models select exactly 939 individuals. However, the queues diverge: 878 participants are shared, while each model exclusively selects 61 individuals.



Important Scientific Scope Note: The 3.0× higher precision (4.92% vs 1.64%) applies *strictly* to the 61-person swapped boundary subgroup, not to overall model precision (which is 5.86% vs 5.64% unweighted). Overall case yield increased from 53 to 55 (+2 potential cases).

BENCHMARK PERFORMANCE & CDC TAYLOR SERIES SURVEY ESTIMATES (20% CAPACITY)

Screening Model	Slots (K)	Unweighted Cases	Unweighted Sens	Survey-Weighted Sens (95% CI)	Survey-Weighted PPV (95% CI)
Model A (Diagnosis Only)	939	53 / 116	45.69%	40.08% [28.9%, 51.2%]	5.79% [3.6%, 8.0%]
Model B (Biomarker Augmented)	**939**	**55 / 116**	**47.41%**	**42.47% [28.4%, 56.5%]**	**6.02% [3.4%, 8.6%]**
Net Change (Δ)	0	**+2 Cases**	**+1.72%**	+2.39% (Overlapping 95% CIs)	+0.23% (Overlapping 95% CIs)

Survey design note: Estimates evaluated using Taylor Series Linearization across masked strata and primary sampling units (WTPH2YR weights, df = 15). Confidence intervals overlap, demonstrating that gains reflect directional operational refinement rather than statistically separated population aggregates.

05 HEALTH EQUITY SUBGROUP AUDIT

Evaluating case detection rates across insurance coverage groups at the 20% budget threshold:

Subgroup	Total Cases	Model A Detection	Model B Detection
Insured Adults	102 cases	51 / 102 (50.0%)	52 / 102 (51.0%)
Uninsured Adults	**13 cases**	**2 / 13 (15.4%)**	**3 / 13 (23.1%)**

Small Denominator Notice: The uninsured evaluation cohort contains only **13 latent cases** in total. Model B identifies 1 additional case (from 2/13 to 3/13). While directionally encouraging, this small sample cannot establish definitive algorithmic fairness or disparity elimination.

06 MULTI-ARCHITECTURE ROBUSTNESS

To evaluate model dependency, we re-evaluated triage under alternative decision tree algorithms:

- **Random Forest:** Model A detects 60 cases; Model B detects 61 cases (+1 case, 77 swaps).
- **HistGradientBoosting:** Model A detects 59 cases; Model B detects 59 cases (0 net gain, 132 swaps).

Confirms that priority shifts occur across architectures, while total yield gains remain modest (+0 to +2 cases).

07 LIMITATIONS & SCOPE

- **Surveillance vs Diagnosis:** Single venous HbA1c ≥ 6.5% represents an elevated screening criterion, not an autonomous clinical diabetes diagnosis.
- **Survey Uncertainty:** Complex survey confidence intervals overlap between models at all budget thresholds.
- **Operational Assumptions:** Model simulations assume 100% testing uptake among prioritized individuals and uniform test costs.

References: ADA Standards of Care (2024) • CDC NCHS NHANES Continuous Biomonitoring (2017–2023) • Obermeyer et al

Pre-specified observational study

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